

1) a) Force on charge @ $x = 8.00 \text{ m}$

$+Q$ @ $x=0$
 $-2Q$ @ $x=5 \text{ m}$
 $+Q$ @ $x=10 \text{ m}$

$$F = k \frac{1Q1Q2}{r^2}, k = 9 \times 10^9$$

Distance: $r = 8.00 - 0 = 8.00 \text{ m} \rightarrow 0.08 \text{ m}$

$F_1 = (9 \times 10^9) \frac{(1.0 \times 10^{-6})^2}{(0.08)^2} = 1.406 \text{ N @ } x=0$ Repulsive, pushes $\rightarrow$

Force on charge @ $x = 5.00 \text{ m}$, $r = 8 - 5 = 3 \text{ m}$, 0.03 m

$F_2 = (9 \times 10^9) \frac{(1.00 \times 10^{-6})(2.00 \times 10^{-6})}{(0.03)^2} = 20.0 \text{ N @ } x=5.00 \text{ m}$ attractive, pushes $\leftarrow$

Force on charge @ $x = 10.00 \text{ m}$, $r = 10 - 8 = 2 \text{ m}$, 0.02 m

$F_3 = (9 \times 10^9) \frac{(1.00 \times 10^{-6})^2}{(0.02)^2} = 22.5 \text{ N @ } x=10.00 \text{ m}$ repulsive, pushes $\rightarrow$

$F_{\text{net}} = 1.406 - 20.0 - 22.5 = -41.1 \text{ N}$

$41.1 \text{ N to the left}$

b) Total electric field @ $x = 11.00 \text{ m}$

$-2Q$ @ $x=0$
 $+Q$ @ $x=5.00 \text{ m}$
 $+3Q$ @ $x=10.00 \text{ m}$
 $-Q$ @ $x=12.00 \text{ m}$

$$E = k \frac{q}{r^2}$$

1. $x=0$, $r = 11 - 0 = 11 \text{ m}$, 0.11 m

$E_1 = (9 \times 10^9) \frac{2.00 \times 10^{-6}}{(0.11)^2} = 1.49 \times 10^6 \text{ N/C}$
Right, toward $-2Q$

2. $x=5$, $r = 11 - 5 = 6 = 0.06 \text{ m}$

$E_2 = (9 \times 10^9) \frac{1.00 \times 10^{-6}}{(0.06)^2} = 2.5 \times 10^6 \text{ N/C}$
Right, away from $+Q$

3. $x=10 \text{ m}$, $r = 11 - 10 = 1 \text{ m} = 0.01 \text{ m}$

$E_3 = (9 \times 10^9) \frac{3.00 \times 10^{-6}}{(0.01)^2} = 27.0 \times 10^6 \text{ N/C}$

Right, away from $+3Q$

4. $x=12.00 \text{ m}$, $r = 12.00 - 11.00 = 1.00 \text{ m} = 0.01 \text{ m}$

$E_4 = (9 \times 10^9) \frac{1.00 \times 10^{-6}}{(0.01)^2} = 9.0 \times 10^6 \text{ N/C}$

Left, toward $-Q$

$E_{\text{net}} = E_{123} - E_4 = 21.99 \times 10^6 \text{ N/C, to the right}$

2. a) The force on q is zero, no net direction

$$b) r = \frac{\text{disanal}}{2} = \frac{5\sqrt{2}}{2} = \frac{10 \times 1.414}{2} = 7.07 \text{ m} = 0.0707 \text{ m}$$

$$K = 9.0 \times 10^9 \text{ N/C}$$

$$q_1 = 7.5 \times 10^{-6} \text{ C}$$

$$q_2 = 2.0 \times 10^{-6} \text{ C}$$

$$r = 0.0707 \text{ m}$$

$$F = (9.0 \times 10^9) \frac{(7.5 \times 10^{-6}) / (2.0 \times 10^{-6})}{(0.0707)^2}$$

$$F = (9.0 \times 10^9) \frac{1.5 \times 10^{-11}}{5.0 \times 10^{-3}} = 0.270 \text{ N}$$

$$F_x = F \cos 45 = 0.270 \frac{1}{\sqrt{2}} = 0.191 \text{ N}$$

$$F_y = F \sin 45 = 0.270 \frac{1}{\sqrt{2}} = 0.191 \text{ N}$$

$$\text{Net force} = 0 \text{ N}$$

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$$KE = eV$$

$$KE = (1.6 \times 10^{-19} \text{ C})(40,000 \text{ V}) = 6.4 \times 10^{-15} \text{ J}$$

$$KE = \frac{1}{2} mv^2$$

$$V = \sqrt{\frac{2KE}{m}}$$

$$V = \sqrt{\frac{2(6.4 \times 10^{-15})}{9.11 \times 10^{-31}}} = \boxed{1.185 \times 10^6 \text{ m/s}}$$

$$4. a) \Delta V = Ed$$

$$\Delta V = (2.50 \times 10^4)(0.0400 \text{ m}) = \boxed{8.00 \times 10^3 \text{ V}}$$

$$b) V(x) = Ex$$

$$x = 1 \text{ m} = 0.01 \text{ m}$$

$$V(0.01) = (2.50 \times 10^4)(0.01) = \boxed{250 \text{ V}}$$

$$c) E = \frac{\Delta V}{d}$$

$$\Delta V = 80 \text{ mV} = 80 \times 10^{-3} \text{ V}$$

$$d = 9 \text{ nm} = 9 \times 10^{-9} \text{ m}$$

$$E = \frac{800 \times 10^{-3}}{9 \times 10^{-9} \text{ m}} = \boxed{8.89 \times 10^6 \text{ V/m}}$$

$$5. U = q \Delta V$$

$$U = 32 \text{ keV} = 32 \times 10^3 \text{ eV}$$

$$q = 2 \times (1.6 \times 10^{-19} \text{ J})$$

$$32 \times 10^3 = 2(1.6 \times 10^{-19}) \Delta V$$

$$\Delta V = 16.0 \times 10^3 \text{ V}$$

$$E = \frac{\Delta V}{d}, \quad d = 2.00 \text{ cm} = 0.0200 \text{ m}$$

$$E = \frac{16 \times 10^3 \text{ V}}{0.0200 \text{ m}} = \boxed{8.00 \times 10^5 \text{ V/m}}$$

$$6. KE = U = \frac{k \cdot q_1 \cdot q_2}{r}$$

$$q_1 = 2e$$

$$q_2 = 79e$$

$$KE = 5 \text{ MeV} = 5.00 \times 10^6 \text{ eV}$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$r = \frac{1.44 \cdot 10^9}{5.00} = \frac{227.52}{5.00} = 45.5 \text{ fm}$$

$$\boxed{4.55 \times 10^{-14} \text{ m}}$$

$$1. C = \frac{Q}{V}$$

$$Q = 3.00 \text{ } \mu\text{C} = 3.00 \times 10^{-6} \text{ C}$$

$$V = 120 \text{ V}$$

$$C = \frac{3.00 \times 10^{-6} \text{ C}}{120 \text{ V}} = 2.50 \times 10^{-8} \text{ F} = \boxed{25.0 \text{ nF}}$$

$$2. a) U = \frac{1}{2} C V^2$$

$$C = 10.0 \text{ } \mu\text{F} = 10.0 \times 10^{-6} \text{ F}$$

$$V = 9.00 \times 10^3 \text{ V}$$

$$U = \frac{1}{2} (10.0 \times 10^{-6} \text{ F}) (9.00 \times 10^3 \text{ V})^2 = \boxed{405 \text{ J}}$$

$$b) V = \sqrt{\frac{2U}{C}}$$

$$U = 40.0 \text{ J}$$

$$C = 8.00 \text{ } \mu\text{F} = 8.00 \times 10^{-6} \text{ F}$$

$$V = \sqrt{\frac{2 \times 40.0 \text{ J}}{8 \times 10^{-6} \text{ F}}} = 10^2 \approx \boxed{3.16 \times 10^3 \text{ V}}$$

$$3. a) C_{\text{total}} = 4 \text{ } \mu\text{F}$$

$$Q_{\text{total}} = C_{\text{total}} V = (10 \text{ } \mu\text{F}) (9.00 \times 10^3 \text{ V}) = 900 \text{ } \mu\text{C}$$

$$Q_{\text{individual}} = \frac{Q_{\text{total}}}{4} = \frac{900 \text{ } \mu\text{C}}{4} = 22.5 \text{ } \mu\text{C}$$

$$\frac{Q_{\text{individual}}}{V} = \frac{22.5 \times 10^{-6} \text{ C}}{9.00 \times 10^3} = 2.50 \text{ } \mu\text{F}$$

$$\text{Capacitance / capacitor} = \boxed{2.50 \text{ } \mu\text{F}} \quad \text{Charge / capacitor} = \boxed{22.5 \text{ } \mu\text{C}}$$

b) parallel connection ensures sufficient capacitance and charge storage at the required voltage

$$U. R_{total} = R + R_{LED}$$

$$R_{total} = 1000 \Omega + 3 \Omega = 1003 \Omega$$

$$V = IR$$

$$I = \frac{V}{R_{total}} = \frac{3.0V}{1003 \Omega} \approx 2.99 \text{ mA}$$

$$b) P = I^2 R$$

$$= (2.99 \times 10^{-3})^2 3 \Omega \approx 26.8 \mu W$$

$$1. a) R_{series} = R_1 + R_2 = 10 \text{ k}\Omega + 5 \text{ k}\Omega = 15 \text{ k}\Omega$$

$$R_3 = \frac{30}{13} \text{ k}\Omega = 2.31 \text{ k}\Omega$$

$$b) I_{series} = \frac{4V}{R_{series}} = \frac{12V}{15 \text{ k}\Omega} = 0.800 \text{ mA}$$

$$c) I_3 = \frac{4V}{R_3} = \frac{12V}{2.31 \text{ k}\Omega} = 5.20 \text{ mA}$$

$$2) a) V_{total} = 1.5V + 1.5V = 3.0V$$

$$I = \frac{V}{R} = \frac{3.0V}{500 \Omega} = 6.0 \text{ mA}$$

$$P = I^2 R = (0.006 A)^2 500 \Omega$$

$$P = 18 \mu W$$

For both circuits: current is 6 mA, power is 18 μW

$$b) R_{internal} = 0.5 \Omega + 0.5 \Omega = 1.0 \Omega$$

$$R_{total} = R + R_{internal} = 500 \Omega + 1.0 \Omega = 501 \Omega$$

$$\left. \begin{array}{l} I = 5.98 \text{ mA} \\ P = 17.9 \mu W \end{array} \right\} \text{ Series}$$

$$\left. \begin{array}{l} I = 5.98 \text{ mA} \\ P = 17.9 \mu W \end{array} \right\} \text{ Parallel}$$

$$P = 17.9 \mu W$$

$$3) a) V_{total} = 4.74V + 1.53V = 6.27V$$

$$R_{internal} = 0.06 \Omega + 0.10 \Omega = 0.16 \Omega$$

$$R_{total} = R_{load} + R_{internal}$$

$$R_{total} = 10.00 \Omega + 0.16 \Omega = 10.16 \Omega$$

$$P = \frac{V_{total}}{R_{total}}$$

$$R_{total}$$

$$I = \frac{6.27V}{10.16 \Omega} \approx 0.6172 A$$

$$b) P = I^2 R_{load}$$

$$= (0.6172 A)^2 10.00 \Omega$$

$$\approx 3.81 W$$

$$c) I = 0.225 A$$

$$R_{internal} = 180 \Omega, (R_{dry cell} = 17.98 \Omega)$$

1. a) $v \rightarrow$
 B out of page

$$F = \downarrow$$

$$(V = \hat{i}, B = \hat{k}, F = -\hat{j})$$

b) $v = \hat{i}$

$B = \rightarrow$

$$F = \downarrow$$

$$(V = \hat{i}, B = \hat{j}, F = -\hat{k})$$

a $\begin{matrix} V \\ \hat{i} \text{ right} \end{matrix}$ $\begin{matrix} B \\ \hat{k} \text{ out} \end{matrix}$ $\begin{matrix} F \\ -\hat{j} \text{ down} \end{matrix}$

b $\begin{matrix} \hat{j} \text{ up} \end{matrix}$ $\begin{matrix} \hat{i} \text{ right} \end{matrix}$ $\begin{matrix} -\hat{k} \text{ in} \end{matrix}$

c) $k = out of page$

$B = \hat{i}$

$$F = \leftarrow$$

$$(V = \hat{k}, B = \hat{j}, F = -\hat{i})$$

c $\begin{matrix} \hat{k} \text{ out} \end{matrix}$ $\begin{matrix} \hat{j} \text{ up} \end{matrix}$ $\begin{matrix} -\hat{i} \text{ left} \end{matrix}$

$$2) \sin \theta = \frac{F}{qVB}$$

$$\sin \theta = \frac{1.40 \times 10^{-6}}{(1.60 \times 10^{-19})(4.00 \times 10^3)(1.25)}$$

$$= 0.175$$

Alternative:

$$\theta = \sin^{-1}(0.175)$$

$$\theta = 10.1$$

$$\theta_1 = 10.1^\circ, \theta_2 = 90^\circ - 10.1^\circ = 79.9^\circ$$

Could be:

$$(10.1^\circ) \text{ or } (79.9^\circ)$$

$$3) a) Q = \frac{(2.66 \times 10^{-26})(5.00 \times 10^6)}{(0.25)(1.20)}$$

$$= \frac{1.33 \times 10^{-19}}{0.272}$$

$$= 4.8 \times 10^{-19} \text{ C}$$

$$b) e = 1.60 \times 10^{-19} \text{ C}$$

$$\frac{q}{e} = \frac{4.8 \times 10^{-19}}{1.60 \times 10^{-19}} = 3, \text{ 3 charge}$$

$$r = \frac{mv}{qB}$$

$$r_{18} = \frac{18}{10} \times r_6$$

$$r_{18} = \frac{18}{10} \times 0.231$$

$$r_{18} = 0.260 \text{ m}$$